

COM668 Verification & Validation

Truth and Worth

Project verification and validation overview

- Verify how true your software is to its specification
- Validate the worth of your software in meeting its operational purpose
- Plan verification and validation at an early stage
- Conduct verification and validation in keeping with your plan
- Evidence both processes have taken place throughout your project
- Report the results gathered in summary and in detail
 - Summarise results in the body of the report
 - Provide detailed (raw) results in an appendix
- Evaluate what reported results mean within the context of your project
 - Discuss significant cases in detail in the body of the report
 - Summarise and discuss broader cases and results in your report

Verification - Specific Goals and the related Specific Practices

- Where the work-product is code, verification is achieved by testing
- However, peer review (or just review) is also a form of walk-through verification
- As a minimum, strive to meet SG3 by performing SP3.1 and 3.2

VERIFICATION

ENGINEERING (ML3)

The purpose of Verification (VER) is to ensure that selected work products meet their specified requirements.

SG 1 Preparation for verification is conducted.

SP 1.1 Select work products to be verified and verification methods to be used.

SP 1.2 Establish and maintain the environment needed to support verification.

SP 1.3 Establish and maintain verification procedures and criteria for the selected work products.

SG 2 Peer reviews are performed on selected work products.

SP 2.1 Prepare for peer reviews of selected work products.

SP 2.2 Conduct peer reviews of selected work products and identify issues resulting from these reviews.

SP 2.3 Analyze data about the preparation, conduct, and results of the peer reviews.

SG 3 Selected work products are verified against their specified requirements.

SP 3.1 Perform verification on selected work products.

SP 3.2 Analyze results of all verification activities.

Validation - Specific Goals and the related Specific Practices

- It is the final product as received by the customer that is validated
- Work-products may pass verification yet fail validation
- As a minimum, strive to meet SG2 by performing SP2.1 and 2.2

VALIDATION

ENGINEERING (ML3)

The purpose of Validation (VAL) is to demonstrate that a product or product component fulfills its intended use when placed in its intended environment.

SG 1 Preparation for validation is conducted.

SP 1.1 Select products and product components to be validated and validation methods to be used.

SP 1.2 Establish and maintain the environment needed to support validation.

SP 1.3 Establish and maintain procedures and criteria for validation.

SG 2 The product or product components are validated to ensure they are suitable for use in their intended operating environment.

SP 2.1 Perform validation on selected products and product components.

SP 2.2 Analyze results of validation activities.

Reporting verification

Test #	Description of Test	Test Data	Requirement tested	Expected Result	Actual Result	Pass/Acceptabel/Fail
1.1	User presented with a login screen to input login credentials	Record on dB of Username and Password	F.01	User able to log in or error detail displayed	User logs in or a generic error is displayed	Acceptable
1.2	Admin able to log out	N.A	F.02	System goes to main public interface	System does nothing	Fail

Project Name:

Test Case Template

Test Case ID: Fun_10

Test Priority (Low/Medium/High): Med

Module Name: Google login screen

Test Title: Verify login with valid username and password

Description: Test the Google login page

Test Designed by: <Name>

Test Designed date: <Date>

Test Executed by: <Name>

Test Execution date: <Date>

Pre-conditions: User has valid username and password

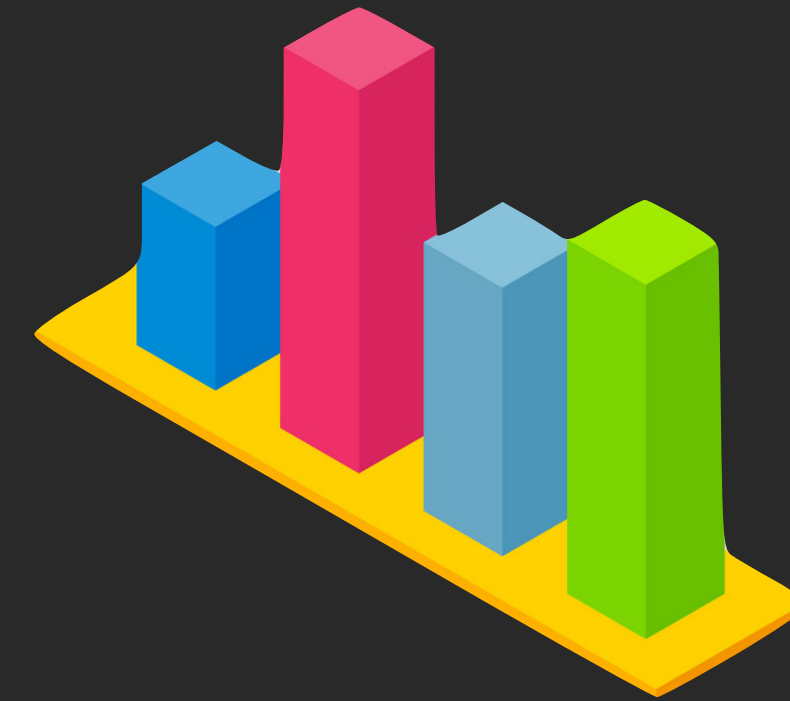
Dependencies:

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Navigate to login page	User= example@gmail.com	User should be able to login	User is navigated to	Pass	
2	Provide valid username	Password: 1234		dashboard with successful		
3	Provide valid password			login		
4	Click on Login button					

Post-conditions:

User is validated with database and successfully login to account. The account session details are logged in database.

Reporting validation



Take the extra step with testing

- Do not just test to meet the reporting requirement
- Test in an active, purposeful rigorous and comprehensive manner
 - Plan continuous testing processes
 - Integrate automated testing into your workflow
 - Specify detailed and comprehensive test cases
 - Show care in the design of survey questions
- Show that testing had an evidenced impacted on your software

COM668 Verification & Validation

Truth and Worth